



WEEK I ASSIGNMENT

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WHAT IS DEEP LEARNING?

Deep learning is a subset of machine learning that uses multilayered neural networks, called deep neural networks, to simulate the complex decision-making power of the human brain. Some form of deep learning powers most of the artificial intelligence applications in our lives today.

My understanding: Even though this is the official definition, I like to think of it as simple as possible. The deep learning is making machines to understand and analyse real world and training them to think like a human and to make them take decisions like a huma.

WHAT IS NEURAL NETWORK?

A neural network is a machine learning program, or model, that makes decisions in a manner similar to the human brain, by using processes that mimic the way biological neurons work together to identify phenomena, weigh options and arrive at conclusions.

My understanding: In simple words it is connection of nodes that work similar to neurons of human brain mimicking the structural and functional components of it.

TYPES OF NEURAL NETWORK:

- Feedforward Neural Networks
- Convolutional Neural Networks (CNNs)
- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory networks (LSTMs)

WHAT IS CNN?

CNN is a type of neural network that is best suited for visual analysis and prediction. This type works very well with images and visuals of any kind.

PIPELINE OF THE PROJECT:

The project pipeline consists of five major steps:

Data collection and
Data loading



Image processing



Building CNN model



Test and Evaluate